

JEE MAIN REVISION MASTER: STATISTICS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Mean, Variance & Standard Deviation	10 PYQs	Dispersion
Standard	Measures of Central Tendency & Median	0 PYQs	Central Tendency

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mean, Variance & Standard Deviation	$\bar{x} = \frac{\sum x_i}{n}, \quad \sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2, \quad \sigma = \sqrt{\text{Variance}}, \quad \text{CV} = \frac{\sigma}{\bar{x}} \times 100$
Measures of Central Tendency & Median	$\text{Empirical: } \text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}, \quad \text{Median} = l + \left(\frac{N/2 - CF}{f} \right) h$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mean, Variance & Standard Deviation (High Yield — 10 PYQs)

Core Principle: Measures of central tendency, variance calculation using mean of squares minus square of mean, and dispersion scaling.

■ **Caution / JEE Pitfall:** Adding a constant c to every observation changes the mean ($\bar{x}' = \bar{x} + c$), but leaves the VARIANCE completely UNCHANGED ($\sigma'^2 = \sigma^2$).

■ **Speed Trick:** If every observation is multiplied by k : new mean = $k\bar{x}$, new standard deviation = $|k|\sigma$, and new variance = $k^2\sigma^2$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The mean and standard deviation of 100 items are 50 and 4, respectively then the sum of all squares of the items is..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "The mean deviation about the mean for the data 4, 7, 8, 9, 10, 12, 13, 17 is..." **[Ans: N/A]**
- **[2024 Shift] Q#3 (MCQ):** "Let a, b, c, d and e be the observations with mean m and standard deviation S . The standard deviation of the observations $a+k, b+k, \dots$ " **[Ans: N/A]**

3.2 Measures of Central Tendency & Median (Standard Yield — 0 PYQs)

Core Principle: Computation of median and mode for discrete and continuous distributions, and Pearson empirical relationship.

■ **Caution / JEE Pitfall:** Sum of absolute deviations is minimum when taken about the MEDIAN ($\sum |x_i - M| = \min$), whereas sum of squared deviations is minimum about the MEAN ($\sum (x_i - \bar{x})^2 = \min$).

■ **Speed Trick:** Mean is heavily sensitive to extreme outlier values, while Median is robust against outliers.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The mean and standard deviation of 100 items are 50 and 4, respectively then the sum of all squares of the items is..." [Ans: N/A]
- **[2025 Shift] Q#2 (MCQ):** "The mean deviation about the mean for the data 4,7,8,9,10,12,13,17 is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The mean and standard deviation of 100 items are 50 and 4, respectively then the sum of all squares of the items is

- (A) 250000
- (B) 251600
- (C) 256100
- (D) 265100

Correct Answer: None

Solution Analysis: For 100 items, the mean is $\bar{x}=50$ and the standard deviation is $\sigma=4$. Using the formula $\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2$ Substitute the given values: $4^2 = \frac{\sum x_i^2}{100} - 50^2$ $16 = \frac{\sum x_i^2}{100} - 2500$ So, $\frac{\sum x_i^2}{100} = 2516$ $\sum x_i^2 = 2516 \times 100 = 251600$ Hence, the sum of all squares of the items is 251600 Correct option: Option B.